

FIITJEE

## ALL INDIA TEST SERIES

### PART TEST – I

**JEE (Main)-2024**

**TEST DATE: 18-11-2023**

---

## ANSWERS, HINTS & SOLUTIONS

### **Physics**

#### PART – A

---

#### SECTION – A

1.

Sol.

B

$$n_1 u_1 = n_2 u_2$$

$$256 \frac{\text{kg}}{\text{m}^3} = n_2 \frac{50\text{g}}{(25\text{cm})^3}$$

$$\Rightarrow 256 \times \frac{1000\text{g}}{(100\text{ cm})^3} = n_2 \frac{50\text{g}}{(25\text{ cm})^3}$$

$$\therefore n_2 = 80 \text{ units}$$

2.

Sol.

A

Time taken when they cross in same direction

$$t_1 = \frac{(60 + 180) / 1000}{90 - 40}$$

When cross in opposite direction,

$$t_2 = \frac{(60 + 180) / 1000}{90 + 40}$$

$$\frac{t_1}{t_2} = \frac{13}{5}$$

3.

C

Sol. Area under a versus x gives  $\frac{v^2 - u^2}{2}$

$$\Rightarrow \frac{v^2 - (\sqrt{10})^2}{2} = 45$$

$$\therefore v = 10 \text{ m/s}$$

4.

C

Sol.

From constraint  $4a_2 = 3a_1$  ... (i)

Where  $a_1$  and  $a_2$  are acceleration of 6 kg and 4 kg blocks respectively

$$4T = 4a_2 \quad \dots (ii)$$

$$66 - 3T = 6a_1 \quad \dots (iii)$$

Solving (i), (ii) and (iii) we get

$$a_1 = 8 \text{ m/s}^2 \text{ and } a_2 = 6 \text{ m/s}^2$$

5.

A

Sol.

$$2Mg = \mu mg$$

$$\Rightarrow M = \frac{\mu m}{2} = \frac{0.4 \times 5}{2} = 1 \text{ kg}$$

6.

A

Sol.

work done by the force is

$$W = \frac{J}{2}(v + u) = \frac{10}{2}(6 + 3) = 45 \text{ J}$$

7.

C

Sol.

$$\frac{1}{2}mu^2 = \frac{1}{2}mv^2 + mg\ell(1 + \cos\theta) \quad \dots (i)$$

$$\Rightarrow v^2 = u^2 - 2g\ell(1 + \cos\theta) \quad \dots (ii)$$

$$\text{Also, } mg\cos\theta = \frac{mv^2}{\ell}$$

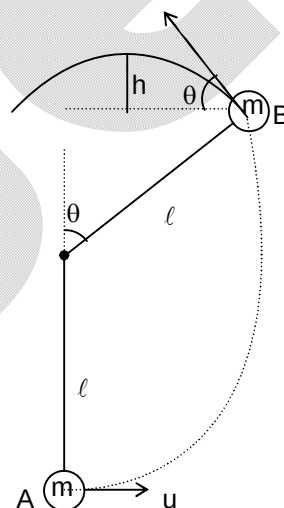
$$v^2 = g\ell\cos\theta$$

From (i) and (ii)

$$\cos\theta = \frac{1}{2}, \theta = 60^\circ$$

$$h = \frac{v^2 \sin^2\theta}{2g} = \frac{3\ell}{16}$$

$$\therefore H = \ell + \ell\cos\theta + \frac{3\ell}{16} = \frac{27}{16}\ell$$



8.

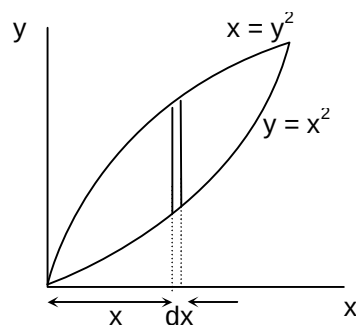
D

Sol.

$$x_{CM} = \frac{\int x dm}{\int dm}$$

$$x_{CM} = \frac{\int_0^1 x \sigma(\sqrt{x} - x^2) dx}{\int_0^1 \sigma(\sqrt{x} - x^2) dx}$$

$$x_{CM} = \frac{9}{20}m$$



9. B  
Sol. FBD of block w.r.t truck

$$f\left(\frac{1.5}{2}\right) = N\left(\frac{0.75}{2}\right) f$$

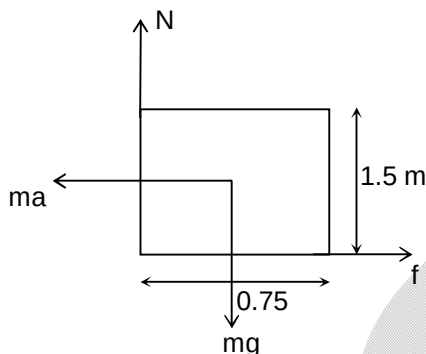
$$N = 2f$$

$$\text{Also, } f = ma = \mu N$$

$$ma = \mu mg$$

$$a = \mu g = 5 \text{ m/s}^2$$

$$\text{Also, } \mu_{\min} = \frac{ma}{mg} = 0.5$$



10. A  
Sol. Angular momentum is conserved about point on the ground.

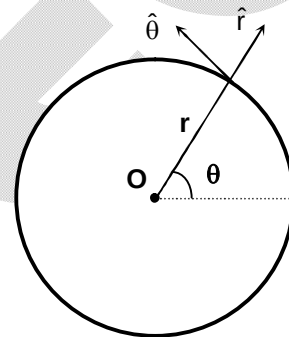
11. D

$$\text{Sol. } \hat{r} = \frac{x\hat{i} + y\hat{j}}{\sqrt{x^2 + y^2}}$$

$$\hat{\theta} = \frac{-y\hat{i} + x\hat{j}}{\sqrt{x^2 + y^2}}$$

$$\therefore \vec{F} = k\hat{\theta}N$$

$$W = k(4\pi R) = 4k\pi R$$



12. C

$$\text{Sol. } f - \frac{3}{5}mg = ma$$

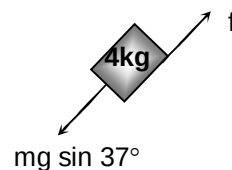
$$\Rightarrow f = m\left(\frac{3g}{5} + a\right)$$

$$\text{Since, } f \leq \mu N = \mu\left(\frac{4}{5}mg\right)$$

$$m\left(\frac{3g}{5} + a\right) \leq \mu\left(\frac{4}{5}mg\right)$$

$$\Rightarrow 8\mu \geq 6 + 2$$

$$\therefore \mu \geq 1$$



13. A

$$\text{Sol. } \vec{P}_f = \frac{mv_0}{2} \hat{j}$$

$$\vec{P}_i = \frac{4}{5}mv_0 \hat{i} + \frac{3}{5}mv_0 \hat{j}$$

Impulse acting on ball

$$\vec{J} = \vec{P}_f - \vec{P}_i$$

$$\therefore \vec{J} = mv_0 \left( -\frac{4}{5} \hat{i} - \frac{1}{10} \hat{j} \right)$$

14. C

Sol. with respect to cart  $\vec{u} = 20\sqrt{3}\hat{j}$

w.r.t. cart  $\vec{a} = -10\sqrt{3}\hat{i} - 10\hat{j}$

minimum radius of curvature

$$\rho_{\min} = \frac{(30)^2}{20} = 45\text{m}$$

15. D

$$\text{Sol. } x_{\text{CM}} = \frac{\frac{\pi R^2}{2}(0) - \frac{\pi R^2}{8}\left(\frac{-R}{2}\right) + \frac{\pi R^2}{8}\left(\frac{-R}{2}\right)}{\frac{\pi R^2}{2} - \frac{\pi R^2}{8} + \frac{\pi R^2}{8}} = \frac{R}{4}$$

$$y_{\text{CM}} = \frac{\frac{\pi R^2}{2}\left(\frac{4R}{3\pi}\right) - \frac{\pi R^2}{8}\left(\frac{2R}{3\pi}\right) + \frac{\pi R^2}{8}\left(\frac{-2R}{3\pi}\right)}{\frac{\pi R^2}{2} - \frac{\pi R^2}{8} + \frac{\pi R^2}{8}} = \frac{R}{\pi}$$

16. A

Sol. For 1<sup>st</sup> collision;

$$mv = mv_1 + 2mv_2$$

$$\Rightarrow v_1 + 2v_2 = v$$

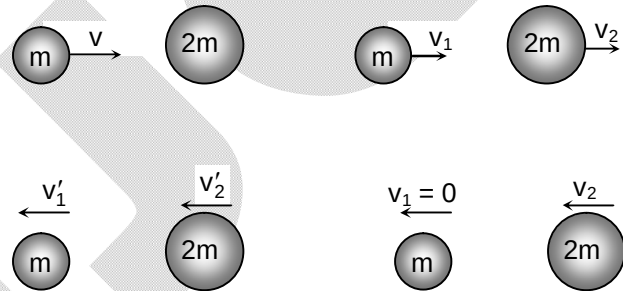
$$\text{and } v_2 - v_1 = v/2$$

$$\therefore v_2 = v/2, v_1 = 0$$

After collision with velocity of ball B is reversed.

Final collision

$$\text{Solving we get } v'_1 = v/2$$



17. D

$$\text{Sol. } [E] = [ML^2T^{-2}]$$

$$[L] = [ML^2T^{-1}]$$

$$[G] = [M^{-1}L^3T^{-2}]$$

$$A = EL^2M^{-5}G^{-2} = [M^0L^0T^0]$$

18. C

$$\text{Sol. } \vec{F} = 2t\hat{i} + 3t^2\hat{j}$$

$$m \frac{d\vec{v}}{dt} = 2t\hat{i} + 3t^2\hat{j}$$

$$\vec{v} = \frac{t^2}{2}\hat{i} + \frac{t^3}{2}\hat{j}$$

$$P = \vec{F} \cdot \vec{v} = (2t\hat{i} + 3t^2\hat{j}) \cdot \left( \frac{t^2}{2}\hat{i} + \frac{t^3}{2}\hat{j} \right)$$

$$P = t^3 + t^5(3/2)$$

$$P = 2^3 + 2^5(3/2) = 8 + 48 = 56 \text{ W}$$

19. B

Sol.  $I = \frac{2}{5}MR^2$

Also,  $\frac{4}{3}\pi R^3 = \pi r^2 t$  ... (i)

$$I = \frac{3}{2}Mr^2 = \frac{2}{5}MR^2$$

$$r = \frac{2R}{\sqrt{15}}$$

From,  $\frac{4}{3}R^3 = \frac{4R^2}{15}t$

$$\therefore t = 5R$$

$$\Rightarrow t = 5R, r = \frac{2R}{\sqrt{15}}$$

20. D

Sol. At  $t = 3.5$  sec

$$h_1 = u(3.5) - \frac{1}{2}(10)(3.5)^2$$

$$h_1 = 3.5u - 61.25$$

Displacement of balloon in 2 sec

$$\Delta h = u(2) + \frac{1}{2}(1)(2)^2 = 2u + 2$$

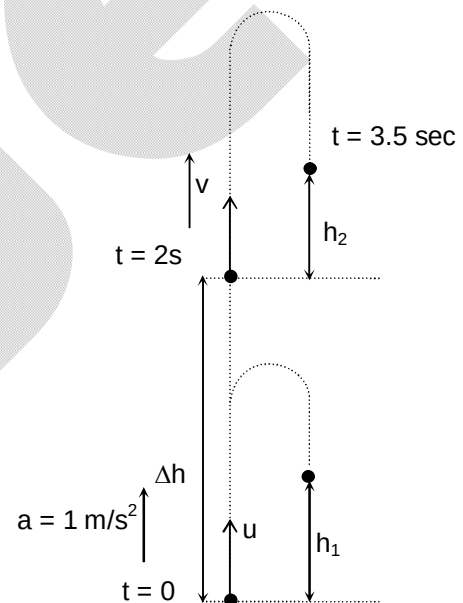
Displacement of stone-2 in 1.5 sec

$$h_2 = (u + 2)(1.5) - \frac{1}{2}(10)(1.5)^2$$

$$= 1.5u + 3 - 11.25$$

Separatin between stones at  $t = 3.5$  sec

$$d = (\Delta h - h_1) + h_2 = 55 \text{ m}$$



### SECTION – B

21. 3

Sol.  $p = \frac{(2)^2(4)}{(4-2)^3} = 2$

$$\frac{dp}{p} = 2 \frac{da}{a} + \frac{db}{b} + 3 \frac{dc}{(4-c)}$$

$$\Rightarrow \frac{dp}{p} \times 100 = 3\%$$

22. 200

Sol. From conservation of mechanical energy

$$\frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} v^2 = \frac{1}{2} k x_m^2$$

$$\Rightarrow \frac{m_1 m_2}{m_1 + m_2} \left( \frac{J}{m_2} \right)^2 = k x_m^2$$

$$\Rightarrow x_m = J \sqrt{\frac{m_1}{m_2(m_1 + m_2)k}} = 24 \sqrt{\frac{2}{4(2+4)(12)}} = 2\text{m} = 200\text{cm}$$

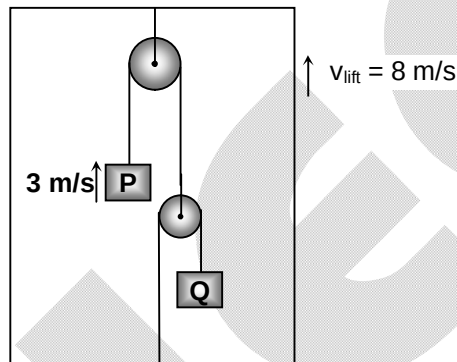
23. 2

Sol. Velocity of relative of lift if

$$v_{QL} = 6 \text{ m/s} (\downarrow)$$

$$v_{QL} = v_Q - v_L$$

$$v_Q = 2 \text{ m/s}$$



24. 3

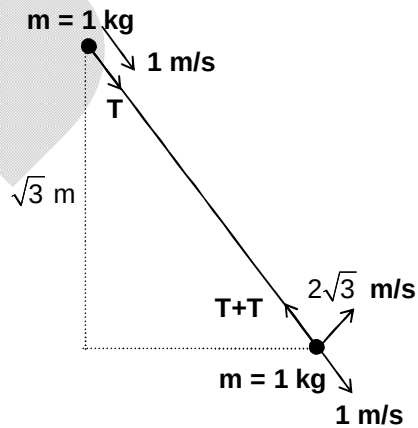
Sol. w.r.t. mass which was at rest initially

$$T + \left( \frac{T}{m} \right) m = \frac{mv^2}{\ell}$$

$$2T = \frac{(1)(2\sqrt{3})^2}{2}$$

$$2T = 6$$

$$T = 3\text{N}$$



25. 8

Sol. Acceleration  $a = \frac{18-15}{3} = 1 \text{ m/s}^2$

For block 1 kg

$$N - 2 - 5 = (1)(1) = 1$$

$$N = 8$$

26. 0

Sol.  $a = 5 \text{ m/s}^2$

$$f = 0$$

27. 2360

Sol. From conservation of mechanical energy,

$$\frac{1}{2} \left[ \frac{m\ell^2}{3} + m \left( \frac{\ell}{2} \right)^2 \right] \omega^2 = \frac{mg\ell}{2} + \frac{mg\ell}{2}$$

$$\Rightarrow \omega^2 = \frac{24g}{7\ell} \quad \dots(i)$$

From conservation of AM about hinge

$$mv_0 \cos 60^\circ \left( \frac{\ell}{2} \right) = \left( \frac{m\ell^2}{3} + \frac{m\ell^2}{4} \right) \omega$$

$$\Rightarrow \omega = \frac{v_0}{4} \quad \dots(ii)$$

From (i) and (ii),

$$\frac{v_0}{4} = \sqrt{\frac{80}{7}}$$

$$\Rightarrow v_0 = 4\sqrt{35} \text{ m/s} = 23.60 \text{ m/s}$$

28. 25

Sol.  $\hat{n} = \frac{\vec{A} \times \vec{B}}{|\vec{A} \times \vec{B}|} = 0.25(\sqrt{3}\hat{i} + \hat{j} - 2\sqrt{3}\hat{k})$

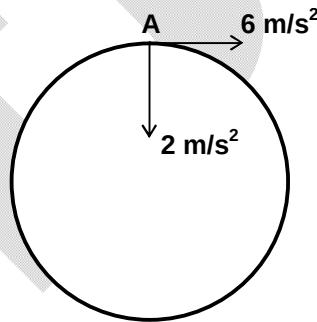
$$\therefore 100\alpha = 100(0.25) = 25$$

29. 40

Sol.  $\omega = 2 \text{ rad/s}$

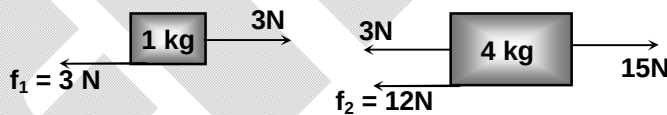
$$a_A = \sqrt{6^2 + 2^2}$$

$$a_A = \sqrt{40} \text{ m/s}^2$$



30. 3

Sol.

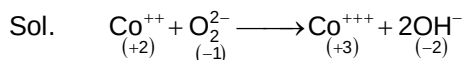


## Chemistry

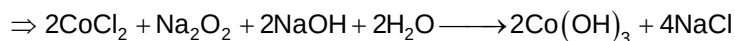
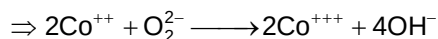
## PART – B

### SECTION – A

31. C

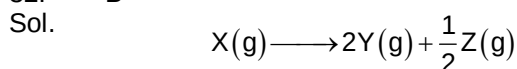


$$\text{Net } \uparrow = 2 \times 1 = 2, \text{ Net } \downarrow = 1 \times 2 = 2$$



$\therefore$  Sum total of stoichiometric coefficients of all the reactants = 2 + 1 + 2 + 2 = 7

32. B



$$\begin{array}{ccc} t = 0, & 600 & - \\ t = t, & 600 - x & 2x \end{array}$$

$$\text{Since, } 600 - x + 2x + \frac{x}{2} = 900$$

$$\Rightarrow 600 + \frac{3x}{2} = 900$$

$$\Rightarrow \frac{3x}{2} = 300 \Rightarrow x = 100 \times 2 = 200$$

$\text{Px(g)}$  undergoes decomposition = 200 mm Hg

$\text{Px(g)}$  original = 600 mm Hg

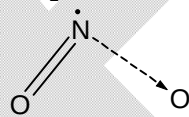
$$\therefore X_{\text{x(g)}} = \frac{200}{600} = \frac{1}{3} = 0.333$$

33. D

Sol. Since red line lies in the visible range which is the Balmer series, so the only possible jump is  $3 \rightarrow 2$ .

34. B

Sol.  $\odot$  Bond order  $\propto \frac{1}{\text{Bond length}}$

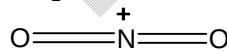


$\equiv$

Odd electron species  $\Rightarrow$  Paramagnetic

$\equiv$

Bent, Bond angle is around  $\cong 134^\circ$

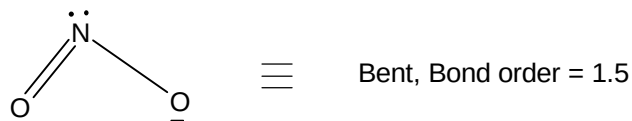


$\equiv$

Linear, Bond order = 2







35. B

Sol.  $\ddot{\text{N}}(\text{SiH}_3)_3 \equiv$  Planar due to backbonding ( $\text{sp}^2$ )

$\ddot{\text{N}}(\text{CH}_3)_3 \equiv$  Pyramidal due to  $\text{sp}^3$  hybridisation and presence of lone pair.

$\ddot{\text{P}}(\text{SiH}_3)_3 \equiv$  Pyramidal since there is no backbonding takes place.

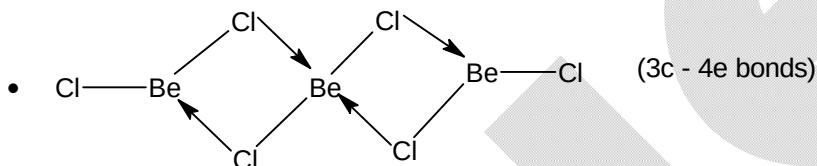
36. D

Sol. Solubility of Gr (2) sulfates

$\text{Be} > \text{Mg} \gg \text{Ca} > \text{Sr} > \text{Ba}$

• Thermal stability of carbonates

$\text{Be} < \text{Mg} < \text{Ca} < \text{Sr} < \text{Ba}$   
 ( $< 100^\circ\text{C}$ ) ( $> 1300^\circ\text{C}$ )



• Solubility of Gr (2) hydroxides increases down the group since decrease in lattice enthalpy is faster than decrease in hydration enthalpy.

37. D

Sol. Chlorine has the highest negative  $\Delta_{\text{eg}}\text{H}_1^\ominus$

- Fluorine has more negative  $\Delta_{\text{eg}}\text{H}_1^\ominus$  as compared to oxygen
- It is difficult to add an electron to the  $-$ vely charged species hence  $\Delta\text{H}_4^\ominus$  is positive.
- $\Delta_{\text{eg}}\text{H}_1^\ominus$  for Cl, F and O are are negative.

38. B

Sol. Sinc  $\text{M}^{4+}(\text{aq})$  and  $[\text{M}(\text{OH})_6]^{2-}(\text{aq})$  ions are present in the solution.

$$\therefore S = [\text{M}^{4+}(\text{aq})] + [\text{M}(\text{OH})_6]^{2-}(\text{aq})$$

$$\Rightarrow S = \frac{K_{\text{sp}}}{[\text{OH}^-]^4} + K_c [\text{OH}^-]^2$$

For minimum solubility  $\frac{dS}{d[\text{OH}^-]} = 0$

$$\Rightarrow -4.K_{\text{sp}}[\text{OH}^-]^{-5} + 2K_c [\text{OH}^-] = 0$$

$$\Rightarrow [\text{OH}^-] \left\{ -4K_{\text{sp}}[\text{OH}^-]^{-6} + 2K_c \right\} = 0$$

Since,  $[\text{OH}^-] \neq 0$

$$\Rightarrow -4K_{\text{sp}}[\text{OH}^-]^{-6} + 2K_c = 0$$

$$\Rightarrow [\text{OH}^-]^{-6} = \frac{K_c}{2K_{sp}}$$

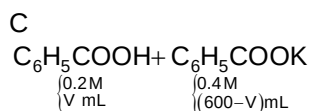
$$\Rightarrow [\text{OH}^-] = \left\{ \frac{K_c}{2K_{sp}} \right\}^{-\frac{1}{6}} = \left\{ \frac{2 \times 10^{-2}}{2 \times 10^{-20}} \right\}^{-\frac{1}{6}}$$

$$\Rightarrow [\text{OH}^-] = 10^{-3} \Rightarrow \text{pOH} = 3$$

$$\therefore \text{pH} = 14 - 3 = 11$$

39.

Sol.



It forms an acidic buffer solution.

$$\therefore \text{pH} = \text{pK}_a + \log \frac{[\text{C}_6\text{H}_5\text{COOK}]}{[\text{C}_6\text{H}_5\text{COOH}]}$$

$$4.5 = 4.2 + \log \frac{0.4(600 - V)}{0.2 \times V}$$

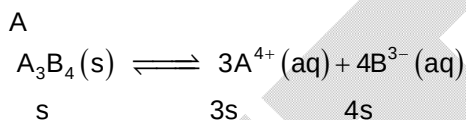
$$0.3 = \log \left( \frac{1200 - 2V}{V} \right)$$

$$\log 2 = \log \left( \frac{1200 - 2V}{V} \right)$$

$$\Rightarrow 2V = 1200 - 2V \Rightarrow V = 300 \text{ mL}$$

40.

Sol.



$$K_{sp} = (3s)^3 (4s)^4$$

$$K_{sp} = 3^3 4^4 (s)^7 \quad \dots (1)$$

$$\text{Now } s = y \text{ g L}^{-1} = \left( \frac{y}{x} \right) \text{ mol L}^{-1} = \left( \frac{y}{x} \right) \text{ M}$$

$$\therefore [\text{A}^{4+}(\text{aq})] = 3s = 3 \left( \frac{y}{x} \right)$$

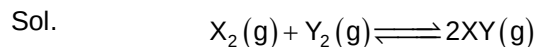
$$K_{sp} = 3^3 4^4 \left( \frac{y}{x} \right)^7$$

$$\text{Now, } \frac{[\text{A}^{4+}(\text{aq})]}{[K_{sp}]} = 3 \cdot \left( \frac{y}{x} \right) / 3^3 4^4 \left( \frac{y}{x} \right)^7$$

$$\Rightarrow \frac{1}{3^2 4^4} \left( \frac{y}{x} \right)^{-6}$$

$$\Rightarrow \frac{1}{9 \times 256} \left( \frac{x}{y} \right)^6 = \frac{1}{2304} \frac{x^6}{y^6}$$

41. A



$$\begin{array}{ccc} t=0 & 2 & 2 & - \\ t=\text{equi.} & \frac{2-x}{8} & \frac{2-x}{8} & \frac{2x}{8} \end{array}$$

$$K_c = \frac{(2x/8)^2}{\left(\frac{2-x}{8}\right)^2} = \frac{4x^2}{(2-x)^2} = 16$$

$$\Rightarrow \frac{2x}{2-x} = 4 \Rightarrow 2x = 8 - 4x$$

$$\Rightarrow x = \frac{4}{3}$$

$$\therefore [XY(g)] = \frac{2 \times 4/3}{8} = \frac{1}{3} = 0.33 \text{ M}$$

42. D

Sol.  $\ln K_{eq} = \left(-\frac{\Delta H^\circ}{R}\right) \frac{1}{T} + \left(\frac{\Delta S^\circ}{R}\right)$

$$\text{Slope } -\frac{\Delta H^\circ}{R} = \frac{4}{6} \quad \text{or } \Delta H^\circ = -\frac{2R}{3}$$

$$\text{y-intercept } \frac{\Delta S^\circ}{R} = 2 \quad \text{or } \Delta S^\circ = 2R$$

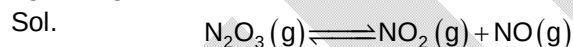
Since,  $\Delta H^\circ < 0$  and  $\Delta S^\circ > 0$

$$\Rightarrow \Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

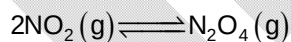
$\Rightarrow \Delta G^\circ$  is always -ve.

$\Rightarrow$  Reaction is always spontaneous

43. C



$$\begin{array}{ccc} \text{Initial} & 4 & - & - \\ \text{Equi.} & 4 - P_1 & P_1 & (P_1 - P_2) \end{array}$$



$$\begin{array}{ccc} \text{Initial} & P_1 & - \\ \text{Equi.} & P_1 - P_2 & P_2 / 2 \end{array}$$

$$\text{Now, } K_{P_1} = \frac{P_1(P_1 - P_2)}{4 - P_1}$$

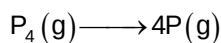
$$\text{Since } K_{P_1} = 1, P_1 = 2$$

$$1 = \frac{2(2 - P_2)}{4 - 2} \Rightarrow P_2 = 1$$

$$\therefore K_{P_2} = \frac{P_2 / 2}{(P_1 - P_2)^2} = \frac{1/2}{(2-1)^2} = 0.50$$

44. C

Sol.



$$\begin{array}{lcl} t = 0, & [Ro] & - \\ t = 60 \text{ min}, & [Ro] - x & 4x \end{array}$$

$$\text{Since } [Ro] - x = 4x \Rightarrow x = \frac{[Ro]}{5}$$

$$\therefore \% \text{ Completion of reaction} = \frac{[Ro]/5}{[Ro]} \times 100 = 20\%$$

$$k = \frac{2.303}{t} \log \frac{[Ro]}{[Ro] - x}$$

$$\frac{0.693}{t_{1/2}} = \frac{2.303}{60} \log \frac{[Ro]}{[Ro] - [Ro]/5}$$

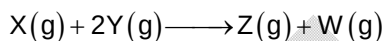
$$t_{1/2} = \frac{60 \times 0.693}{2.303 \times (\log 5 - \log 4)} = 180 \text{ min.}$$

For 1<sup>st</sup> order reactions, rate decreases exponentially with time.

For  $[P_4(g)] = [P(g)]$ , at  $t = 60 \text{ min}$ . 20% reaction completes, so no effect of the initial concentration of  $P_4(g)$ .

45. C

Sol.



$$\begin{array}{lclcl} t = 0, & 0.6 & 0.8 & - & - \\ t = t, & 0.6 - x & 0.8 - 2x & x & x \end{array}$$

$$\text{Since, } 0.8 - 2x = 0.2 \Rightarrow x = 0.3$$

$$\therefore t = t \quad P_{X(g)} = 0.3 \text{ atm}, \quad P_{Y(g)} = 0.2 \text{ atm}$$

Since, reaction is elementary

$$r = k[X]^1[Y]^2$$

$$\therefore \frac{r_t}{r_o} = \frac{k(0.3)(0.2)^2}{k(0.6)^2(0.8)^2} = \frac{3 \times 2 \times 2}{6 \times 8 \times 8} = \frac{1}{32}$$

46. C

Sol.

$$\text{Mosely equation } \sqrt{v} = \sqrt{\frac{C}{\lambda}} = a(Z - b)$$

$$\therefore \lambda \propto \frac{1}{(Z - b)^2}$$

For  $K_\alpha$  line  $b = 1$

$$\Rightarrow \frac{\lambda_2}{\lambda_1} = \frac{(Z_1 - 1)^2}{(Z_2 - 1)^2} \Rightarrow \frac{\lambda_2}{\lambda} = \frac{(57 - 1)^2}{(29 - 1)^2}$$

$$\Rightarrow \frac{\lambda_2}{\lambda} = \frac{56 \times 56}{28 \times 28} \Rightarrow \lambda_2 = 4\lambda$$

47. A

Sol. Since  $\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$

$$\Rightarrow P + \frac{a}{V_m^2} = \frac{RT}{V_m - b}$$

$$\Rightarrow PV_m = RT \left( \frac{V_m}{V_m - b} \right) - \frac{a}{V_m}$$

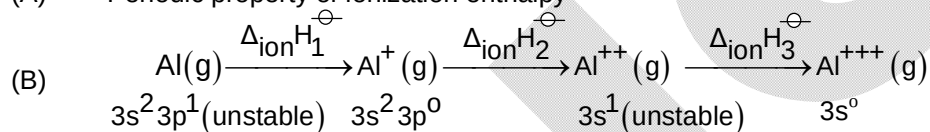
$$\Rightarrow \frac{PV_m}{RT} = \frac{V_m}{V_m - b} - \frac{a}{V_m RT}$$

$$\Rightarrow Z = \frac{V_m}{V_m - b} - \frac{a}{V_m R \left( \frac{a}{Rb} \right)}$$

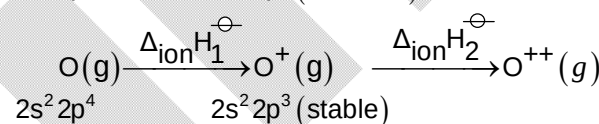
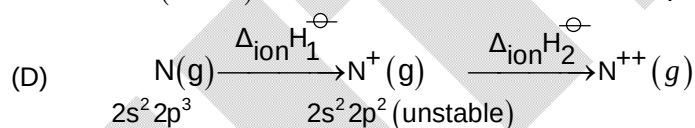
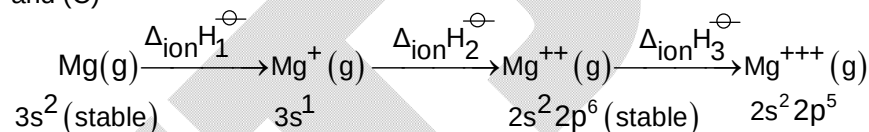
$$\Rightarrow Z = \frac{V_m}{V_m - b} - \frac{b}{V_m} = 1 + \frac{b^2}{V_m(V_m - b)}$$

48. D

Sol. (A) Periodic property of ionization enthalpy



and (C)



49. B

Sol.  $\text{PF}_3 \equiv (2p - 3d)$

$\text{BF}_3 \equiv (2p - 2p) \rightarrow$  Maximum extent of backbonding.

$(\text{SiH}_3)_3 \text{N} \equiv (2p - 3d)$

$(\text{SiH}_3)_2 \text{O} \equiv (2p - 3d)$

50. D

Sol. Both  $\text{BeO}$  and  $\text{Al}_2\text{O}_3$  react with acids and bases.

- Both  $\text{Be}_2\text{C}$  and  $\text{Al}_4\text{C}_3$  on yields  $\text{CH}_4$  on hydrolysis
- Both Be and Al can form complexes
- $\text{BeH}_2$  (Solid)  $\equiv$  hexagonal crystalline form
- $\text{AlH}_3$  (Solid)  $\equiv$  octahedral form

SECTION – B

51. 4

Sol.  $204 \text{ eV} = 13.6 \left( \frac{1}{1^2} - \frac{1}{(2n)^2} \right) \cdot Z^2 \quad \dots (1)$

$40.8 \text{ eV} = 13.6 \left( \frac{1}{n^2} - \frac{1}{(2n)^2} \right) \cdot Z^2 \quad \dots (2)$

From (1) and (2)

$$\frac{204}{40.8} = 1 - \frac{1}{4n^2} / \frac{1}{n^2} - \frac{1}{4n^2}$$

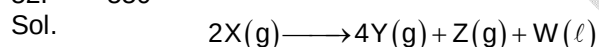
$$5 = \frac{4n^2 - 1}{3} \Rightarrow n^2 = 4 \Rightarrow n = 2$$

From (1),  $204 = 13.6 \left( 1 - \frac{1}{16} \right) Z^2$

$$204 = \frac{13.6 \times 15}{16} Z^2$$

$$\Rightarrow 16 \times 204 = 204 \times Z^2 \Rightarrow Z = 4$$

52. 550



|        |                    |                 |                    |    |
|--------|--------------------|-----------------|--------------------|----|
| t = 0  | X <sub>0</sub>     | –               | –                  | –  |
| t = 30 | X <sub>0</sub> – x | 2x              | x / 2              | 25 |
| t = 60 | X <sub>0</sub> – y | 2y              | y / 2              | 25 |
| t = ∞  | –                  | 2X <sub>0</sub> | X <sub>0</sub> / 2 | 25 |

Now, at t = ∞

$$825 = 2X_0 + \frac{X_0}{2} + 25$$

$$\Rightarrow 800 = \frac{5X_0}{2} \Rightarrow X_0 = 320 \text{ mm.}$$

Now, at t = 30

$$320 - x + 2x + x / 2 + 25 = 645$$

$$3x / 2 + 345 = 645 \Rightarrow x = 200 \text{ mm}$$

$$\text{Now, } K = \frac{1}{30} \ln \frac{X_0}{X_0 - x} = \frac{1}{60} \ln \frac{X_0}{X_0 - y}$$

$$\Rightarrow 2 \ln \frac{320}{320 - 200} = \ln \frac{320}{320 - y}$$

$$\Rightarrow \left( \frac{320}{120} \right)^2 = \frac{320}{320 - y}$$

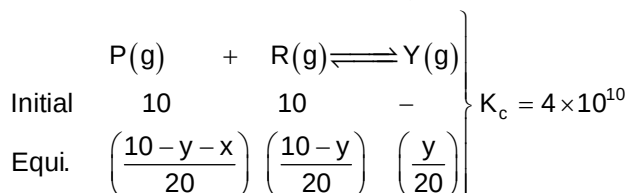
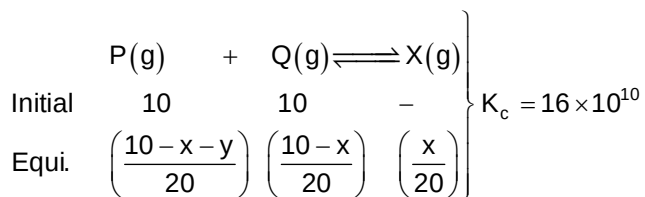
$$\Rightarrow \frac{320 \times 320}{120 \times 120} = \frac{320}{320 - y}$$

$$\Rightarrow 320 - y = 45 \Rightarrow y = 275 \text{ mm}$$

$$\therefore Y_{(P_{60})} = 2y = 2 \times 275 = 550 \text{ mm}$$

53. 50

Sol.



Since,  $K_c$  of both reactions are very large.

So,  $x + y \approx 10$  ... (1)

Now,  $\frac{16 \times 10^{10}}{4 \times 10^{10}} = \frac{x}{20} \times \frac{20^2}{(10-x-y)(10-x)} \bigg/ \frac{y}{20} \times \frac{20^2}{(10-x-y)(10-y)}$

$4 = \frac{x(10-y)}{y(10-x)}$  ... (2)

Solving (1) and (2)

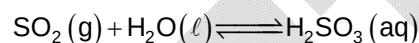
$x = \frac{20}{3}, y = \frac{10}{3}$

$\therefore [Q]_{\text{equi}} = n = 10 - x = 10 - \frac{20}{3} = \frac{10}{3}$

$\therefore 15n = 15 \times \frac{10}{3} = 50$

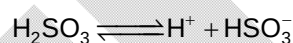
54. 2

Sol.



$(1.28 \text{ gL}^{-1}) \quad (0.02 \text{ M})$

$= \frac{1.28}{64} \text{ mol L}^{-1} = 0.02 \text{ M}$



$0.02 \text{ M} \quad - \quad -$

$0.02 - x \quad x \quad x$

Now  $K_{a_1} = \frac{x^2}{0.02 - x}$

$10^{-2} = \frac{x^2}{0.02 - x} \text{ or } x^2 + (10^{-2})x - 2 \times 10^{-4} = 0$

On solving  $x = 10^{-2} \Rightarrow [\text{H}^+] = 10^{-2} \text{ M}$

$\Rightarrow \text{pH} = -\log(10^{-2}) = 2.00$

55. 33

Sol.

$\left( P + \frac{x}{V_m} \right) (V_m - y) = RT$

$$\Rightarrow P = \frac{RT}{V_m - y} - \frac{x}{V_m}$$

At critical point  $\frac{dP}{dV_m} = 0$  and  $\frac{d^2P}{dV_m^2} = 0$

Or  $\frac{x}{V_m^2} = \frac{RT}{(V_m - y)^2} \quad \dots (1)$

$$\frac{d^2P}{dV_m^2} = + \frac{2RT}{(V_m - y)^3} - \frac{2x}{V_m^3} = 0$$

or  $\frac{x}{V_m^3} = \frac{RT}{(V_m - y)^3} \quad \dots (2)$

Dividing (2) by (1)

$$V_m = V_m - y \Rightarrow y = 0$$

Which is NOT possible.

So, gas has no critical point.

$\therefore$  Answer is 33.

56. 7

Sol. Except  $\text{Me}_2\text{O}$ ,  $\text{MeNCS}$  and  $\text{Me}_3\text{N}$  all other species will show backbonding.

57. 2

Sol. Planar (x)  $\equiv \text{NO}_3^-$ ,  $\text{BrF}_3$ ,  $\text{XeF}_2$ ,  $\text{XeF}_4$ ,  $\text{CH}_3^+$ ,  $\text{ICl}_2^+$

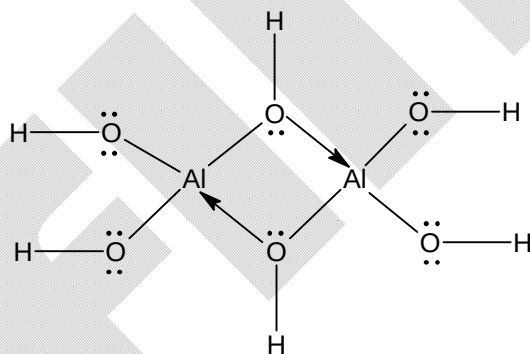
Non-planar (y)  $\equiv \text{XeO}_2\text{F}_2$ ,  $\text{SF}_4$ ,  $\text{PCl}_4^+$ ,  $\text{PCl}_5$ ,  $\text{IF}_5$ ,  $\text{SbF}_4^+$ ,  $\text{SeF}_3$ ,  $\text{SF}_5^-$

$$x = 6, \quad y = 8$$

$$|x - y| = |6 - 8| = 2$$

58. 12

Sol. Dimer of  $\text{Al}(\text{OH})_3 \equiv \text{Al}_2(\text{OH})_6$



$$(2c - 2e) = 10$$

$$(3c - 4e) = 2$$

$$\text{Total} = 12$$

59. 18

Sol.  $Z = 46$ , the electronic configuration is

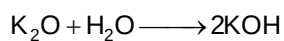
$$1s^2 2s^2 2p^6 \quad 3s^2 3p^6 3d^{10} \quad 4s^2 4p^6 4d^{10}$$

$$\left. \begin{array}{l} s - \text{electrons}(m_\ell = 0) \equiv 8 \\ p - \text{electrons}(m_\ell = 0) \equiv 6 \\ d - \text{electrons}(m_\ell = 0) \equiv 4 \end{array} \right\} \text{Total } 18e^- (s)$$

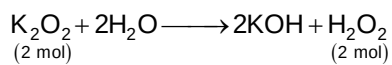


60. 3

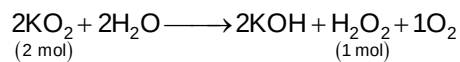
Sol.



$$n_{\text{H}_2\text{O}_2} = 0$$



$$n_{\text{H}_2\text{O}_2} = 2$$



$$n_{\text{H}_2\text{O}_2} = 1$$

$$\text{Total} = 0 + 2 + 1 = 3 \text{ mol of H}_2\text{O}_2$$

# Mathematics

## PART – C

### SECTION – A

61.

B

Sol.

$$p(x) = x^2 + 4x + 5$$

$$\lim_{x \rightarrow 1} f(x) = e^{3/10}$$

62.

C

Sol.

$$g(x) + g'(x) = 12x^2 + 2$$

63.

C

Sol.

$$I = \int_{-2}^2 (-x)^3 \ln \left( \frac{15^x + 5^x + 3^x + 1^x}{15^x} \right) dx$$

$$2I = \int_{-2}^2 x^4 \ln 15 dx$$

64.

A

Sol.

$$f(x) = 25 \Rightarrow x = 3$$

$$\left. \frac{d}{dx} f^{-1}(x) \right|_{x=25} = \left. \frac{1}{f'(x)} \right|_{x=3}$$

65.

C

Sol.

$$e^{x^4} \leq f(x) \leq e^{x^2} \quad \forall x \in (0, 1)$$

$$e^{x^2} \leq f(x) \leq e^{x^4} \quad \forall x \in (1, \infty)$$

66.

B

Sol.

$$I_2 = \frac{1}{4e} \int_0^1 \frac{e^t}{1+t} dt$$

67.

D

Sol.

$$\frac{dy}{dx} = \frac{6y}{x^3}$$

$$\ln|y| = \frac{-3}{x^2} + c$$

$$x^2 \ln|y| = 3(x^2 - 1)$$

68.

D

Sol.

$$\frac{\pi}{4} ab - \frac{1}{2} ab$$

where a, b are length of semi minor and major axis.

69.

A

Sol.

$$\text{Required area } A = \int_0^{\frac{a}{1+a^2}} \left( x - ax^2 - \frac{x^2}{a} \right) dx = \frac{1}{6} \left( \frac{a^2}{(1+a^2)^2} \right)$$

70. B

Sol.  $\lim_{x \rightarrow 0} \frac{\sin 2x - \cos 3x + ke^{2x} - \frac{x^3}{3}}{x^n}$   
 $k = 1$   
 $L = 4$

71. A

Sol.  $f(x) = \frac{e^x(x-1)+2}{x^2}$

72. A

Sol. Conceptual

73. D

Sol. Equation of tangent  
 $y = 4ex - 2e$

74. A

Sol.  $f(x) = \frac{1}{4x} + \frac{3x^3}{4}$

75. D

Sol.  $f(x) = \frac{x-2}{x^2+1}$

76. B

Sol.  $I = \frac{1}{2023}$

77. C

Sol.  $f'(x) = e^{x(x-1)} \cdot (x+1)(x+2)$

78. B

Sol.  $f'(x) = 0$  at  $-1, 3$

79. A

Sol. Conceptual

80. C

Sol.  $f(x) = 2x^3 + 2x^2 - x + 2$

## SECTION – B

81. 1

Sol.  $\int_0^1 f(r-1+x) dx = \int_0^1 f(r-x) dx$

82. 85

Sol. Replace  $x \rightarrow 2-x$

83. 18

Sol.  $\int_{-a}^a (\{x\} + \{1-x\}) dx = 18$

84. 3

Sol. Equation of tangent  $x + 2 = 0$

85. 9

Sol.  $k = 9$

$$a = -\frac{5}{2}$$

86. 9

Sol.  $f(x) = 3^x$

87. 3

Sol. Conceptual

88. 70

Sol.  $f'(x) = ax^2 + bx + 5 > 0$

89. 4

Sol. Conceptual

90. 6

Sol.  $f(x) = -3\left(x + \frac{1}{x}\right)$